Vipul Sapra

Kharghar, Navi Mumbai | 8888-033-168 | vpsapra5@gmail.com | linkedin.com/in/vipul-sapra-ab30a09a

EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Python Programming, Advanced SQL (CTEs, Window Functions, Stored Procedures), Object-Oriented Programming, Exception Handling, REST APIs, Data Reconciliation, ETL & Data Pipelines, Anomaly Detection, Threshold Tuning, Predictive Modeling, SARIMA Time-Series, Data Governance, Agile Delivery

Programming Languages & Tools: Python, Pandas, PySpark, Azure (ADF, Databricks, Synapse, Azure SQL), AWS (S3, Redshift), Snowflake, Apache Spark, Power BI, Tableau, Git, Collibra, KIRO IDE, SQL Server, PostgreSQL

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Software & AI Systems Engineer - Python, Advanced SQL & Cloud Platforms — **May 2022 – Present**

- **Engineered autonomous agentic AI workflows** using KIRO IDE, prompt orchestration, and custom Python scripts to automate schema discovery, data profiling, and regex classification, reducing manual software delivery turnaround by 40-50%.
- **Developed advanced Python analytical pipelines** and complex SQL queries leveraging window functions, CTEs, and optimization techniques for high-throughput reconciliation, data quality firewalls, and data-intensive compliance workflows.
- **Architected distributed cloud data platforms** utilizing Azure (ADF, Databricks, Synapse, Azure SQL) and AWS (S3, Redshift) to process hundreds of terabytes of structured and unstructured datasets with 99.9% availability SLAs.
- **Deployed autonomous machine learning anomaly detection** models and XGBoost classification workflows identifying distribution drift and ETL discrepancies with 95% precision, ensuring rigorous pipeline integrity and risk mitigation.
- **Engineered automated testing and evaluation** frameworks for conversational AI platforms using Python, cutting manual QA turnaround by 50% and accelerating secure production release cycles in Agile delivery teams.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Built predictive machine learning models** using Scikit-Learn, XGBoost, and Random Forests to forecast customer behavior and deliver actionable signals that improved operational service reliability by 20%.
- **Architected advanced SARIMA time-series models** to predict field support and demand patterns, optimizing resource allocation, threshold tuning, and response times by 20%.
- **Integrated PySpark and distributed data processing** workflows to handle large-scale financial and operational telemetry, applying rigorous debugging, unit testing, and version control.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Designed executive Key Operating Metric dashboards** and semantic models using Power BI (DAX) and Tableau to translate complex transactional data into strategic insights.
- **Implemented end-to-end data lineage governance** using Collibra and automated reconciliation firewalls, reducing operational investigation turnaround by over 80%.
- **Collaborated with cross-functional stakeholders** to translate business compliance requirements into robust dimensional models (Star/Snowflake schemas, SCD Type 1/2).

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.